THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 07

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 units	$\sqrt{(4-(-4))^2+(3-(-3))^2}=\sqrt{64+36}=10$
2	**(a)** 13 units	$\sqrt{25+144}=13$
3	**(c)** $(0,\,0)$	intersection of the axes
4	**(d)** $5x-8$	degree 1
5	**(a)** 13	$3(4)+(1)=13$
6	**(a)** terminating	denominator 25 is of the form 2^a5^b
7	**(b)** $\frac{1}{7}$	$\frac{1}{7}=0.\overline{142857}$
8	**(c)** infinitely many rational numbers	density of $\mathbb{Q}$
9	**(d)** $9x^2-24x+16$	$(a-b)^2=a^2-2ab+b^2$
10	**(d)** $x^2-9x+18$	$(x+a)(x+b)=x^2+(a+b)x+ab$
11	**(c)** 75°	angle at the centre is double the angle on the circle
12	**(b)** 8 cm	$d=\sqrt{10^2-6^2}=8$
13	**(b)** 88°	opposite angles of a cyclic quadrilateral add to 180°
14	**(d)** 12 sq. units	$s=7$; $\sqrt{(s-a)(s-b)(s-c)(s-d)}=12$
15	**(b)** 44 cm	$\frac{180}{360}\times 2\pi r=44$
16	**(c)** 35 cm^2	area $=\frac{1}{2}bh$
17	**(d)** Jñān-Chaupaḍ	as given in the chapter
18	**(b)** $\frac{3}{5}$	9 favourable out of 15 equally likely balls
19	**(c)** 47	substitute $n=10$
20	**(d)** 64	$ar^{n-1}=1\times(-2)^6=64$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Distance of $(-5,-12)$ is $\sqrt{25+144}=13$; distance of $(13,0)$ is 13. Yes, both lie on the circle of radius 13 centred at the origin.	1+1
22	$x+(x+10)=64\Rightarrow x=27$; numbers 27 and 37.	1+1
23	$-850+1200-450=-100$: he is in debt of ₹100.	1+1
24	$55^2=60\times50+25=3025$; $65^2=70\times60+25=4225$.	1+1
25	$2\times50^\circ=100^\circ$.	2
26	Arc $=\frac{1}{4}\times2\times\frac{22}{7}\times14=22$ cm; perimeter $=22+28=50$ cm.	1+1
27	Multiples of 3: 3, 6 ⇒ $\frac{2}{8}=\frac{1}{4}$.	1+1
28	$21-3(n-1)=-81\Rightarrow n=35$ (1). $21-3(n-1)=0\Rightarrow n=8$, so 0 is the 8th term (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Fourth vertex $(2,5)$. Diagonal $=\sqrt{36+16}=2\sqrt{13}$ units.	1+2
30	$3x=15\Rightarrow x=5$; check $3(5)-4=11$ ✓. Example: the sum of two numbers is 64 and one is 10 more than the other: $x+(x+10)=64\Rightarrow 2x+10=64$.	1+1+1
31	Multiply numerator and denominator by 5^2 to get 10^3 in the denominator ⇒ exactly 3 decimal places (the last digit is non-zero).	1+2
32	$x(x-4)=96\Rightarrow x^2-4x-96=0\Rightarrow(x-12)(x+8)=0$. $x=12$ m (reject −8); breadth 8 m.	1+1+1
33	Diagonal BD splits the parallelogram into two congruent triangles (1), so $\text{ar}(ABD)=42$ cm^2 (2).	1+2

34	For few trials the relative frequency can differ much from the theoretical value, but as the number of trials grows it tends to get closer to the theoretical probability (e.g. tosses of a fair coin approach 0.5).	3
35	At 6 cm: $2\sqrt{100-36}=16$ cm; at 8 cm: $2\sqrt{100-64}=12$ cm (2). $16>12$ and $6<8$: the chord nearer the centre is longer (1).	2+1
	OR: Let chords AB, CD (equal) meet at P and draw $OM\perp AB, ON\perp CD$; $OM=ON$ (equal chords) (1). Right triangles OMP, ONP are congruent (RHS) ⇒ $PM=PN$ (1). Since $AM=CN$ (halves of equal chords), $AP=CP$ and so $PB=PD$ (1).	1+1+1
36	$3n-7=332\Rightarrow n=113$ (1). $3n-7=557\Rightarrow 3n=564\Rightarrow n=188$, a natural number, so 557 is the 188th term (2).	1+2
	OR: $a+2d=12$, $a+49d=106$ ⇒ $47d=94$, $d=2$, $a=8$ (2). $t_{29}=8+28\times 2=64$ (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\sqrt{2}$. (ii) Draw $OA=1$, perpendicular $AB=1$, then arc of radius OB meets the line at $P=\sqrt{2}$. (iii) Proof by contradiction shows $\sqrt{2}$ is irrational. (iv) $1.4142\ldots$ non-terminating, non-repeating.	1+2+1+1
	OR: (i) $80=2^4\times 5$ ⇒ $\frac{13}{80}$ terminates; $12=2^2\times 3$ and $343=7^3$ ⇒ repeating. (ii) $\frac{13}{80}=\frac{13\times 125}{10^4}=0.1625$. (iii) $\frac{5}{12}=0.41666\ldots=0.41\overline{6}$; repeating digit 6.	2+1+2
38	(i) 14 m path: $\sqrt{25^2-7^2}=24$ m. 48 m path: $\sqrt{25^2-24^2}=7$ m (2). (ii) The 48 m path (longer chord is nearer) (1). (iii) $24+7=31$ m (1). (iv) $24-7=17$ m (1).	2+1+1+1
	OR: Let ABCD be cyclic with centre O. Arc BCD: A is outside it, so $\angle BAD=\frac{1}{2}$(reflex $\angle BOD$) (2). Arc BAD: C is outside it, so $\angle BCD=\frac{1}{2}\angle BOD$ (2). Sum $=\frac{1}{2}\times 360^\circ=180^\circ$ (1).	2+2+1
39	(i) $\frac{22}{7}\times 1225=3850$ m^2 (1). (ii) $\frac{22}{7}\times 49=154$ m^2 (1). (iii) $3850-154=3696$ m^2 (1). (iv) $2\pi(35)+2\pi(7)=220+44=264$ m (2).	1+1+1+2
	OR: (i) $AB=AC=BC=r$ (radii) (1). (ii) $\angle CAB=\angle BAD=60^\circ$, so $\angle CAD=120^\circ$ (2). (iii) Each outer arc subtends 240° at its centre: total $=2\times\frac{240}{360}\times 2\pi r=\frac{8\pi r}{3}$ (2).	1+2+2
40	(i) Points $(1,1),(2,5),(3,9),(4,13),(5,17)$ lie on the line $y=4x-3$ (2). (ii) $(1,3),(2,6),(3,12),(4,24),(5,48)$ do not lie on a line; slopes 3, 6, 12, 24 keep rising (2). (iii) An AP grows by a fixed amount (linear), a GP by a fixed multiple (much faster) (1).	2+2+1
	OR: (i) $S_{58}-S_{24}=29\times 59-12\times 25=1711-300=1411$ (2). (ii) 34 terms; $\frac{25+58}{2}\times 34=1411$ ✓ (2). (iii) Āryabhaṭa, in the Āryabhaṭīya, Chapter 2, verse 19 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.